

SHETH L.U.J AND SIR M.V COLLEGE

The screenshot shows the RStudio interface with the following components:

- Console:** Displays the execution of R code for loading libraries, reading data, identifying columns, categorizing data, and running a Chi-square test.
- Environment:** Lists the objects in the environment, including 'cat_col', 'cat_cols', 'col', 'heights', 'malaria_col', 'med_val', 'num_col', 'num_cols', and 'table_data'.
- Files:** Shows the file explorer with various files and folders, including 'animal.xlsx', 'animex1.csv', 'annual-number-of-deaths-by-cause.c', 'creditcard.csv', 'creditcard.xlsx', 'crimes-against-women-2001-2014.csv', 'hackathon', 'korean_drama.csv', 'p1_S120.R', 'p1.R', 'p2_S120.R', 'p3_S120.R', 'p4_S120.R', 'p6_S120.R', 'p7_S120.R', and 'no extra'.

```
> # --- 1. Load Libraries ---
> library(stats)
>
> # --- 2. Load and Prepare Data ---
> deaths_df <- read.csv("C:/Users/yukta/Downloads/annual-number-of-deaths-by-cause.csv")
>
> # 2a. Identify the column for Malaria deaths
> malaria_col <- "Deaths...Malaria...Sex...Both...Age...All.Ages..Number.."
>
> # 2b. Categorize Malaria deaths into 'High' or 'Low' based on the median
> med_val <- median(deaths_df[[malaria_col]], na.rm = TRUE)
> deaths_df$malaria_Level <- ifelse(deaths_df[[malaria_col]] > med_val, "High", "Low")
>
> # 2c. Categorize Year into two time periods
> deaths_df$Period <- ifelse(deaths_df$Year < 2005, "Early (Pre-2005)", "Late (Post-2005)")
>
> # --- 3. Create Contingency Table and Run Chi-square ---
> table_data <- table(deaths_df$malaria_Level, deaths_df$Period)
> chi_sq_result <- chisq.test(table_data)
>
> # --- 4. Result & Explanation ---
> print(chi_sq_result)

Pearson's Chi-squared test with Yates' continuity correction

data: table_data
X-squared = 28.122, df = 1, p-value = 1.139e-07

> print("Yukta Sonawane S120")
[1] "Yukta Sonawane S120"
>
```